

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271888

Kind Code

A1

Publication Date

August 28, 2025

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CURRENT GENERATOR FOR IDAC FOR CORNER INDEPENDENCE

Abstract

Apparatuses, systems, and methods for improved current generators for IDAC corner independence are provided. An exemplary apparatus may include current generator circuitry and a IDAC. The current generator circuitry is configured to generate a reference current. The IDAC is configured to generate an IDAC current based at least on the reference current. The current generator circuitry may adjust or update the reference current based on a change in temperature and/or process. This results in an adjustment or update in the IDAC current.

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Family ID: 1000007759133

Appl. No.: 18/588989

Filed: February 27, 2024

Publication Classification

Int. Cl.: G05F3/24 (20060101); H03M1/74 (20060101)

U.S. Cl.:

CPC G05F3/242 (20130101); H03M1/74 (20130101);

Background/Summary

TECHNOLOGICAL FIELD

[0001] Example embodiments of the present disclosure relate generally to a current generator for a

current digital-to-analog converter (IDAC), particular to systems, apparatuses, and methods for a current generator for IDAC corner independence.

BACKGROUND

[0002] A current digital-to-analog converter (IDAC) may source or sink a current that may be output. The output of IDACs may, however, be corner dependent. For example, the output may vary due to variations in temperature or process.

[0003] The inventors have identified numerous areas of improvement in the existing technologies and processes, which are the subjects of embodiments described herein. Through applied effort, ingenuity, and innovation, many of these deficiencies, challenges, and problems have been solved by developing solutions that are included in embodiments of the present disclosure, some examples of which are described in detail herein.

BRIEF SUMMARY

[0004] Various embodiments described herein relate to current generators for IDACs.

[0005] In accordance with some embodiments of the present disclosure, an example apparatus is provided. The apparatus may comprise: a current generator circuitry comprising: an op amp comprising a first input, a second input, and an output, wherein the first input is electrically connected to a reference voltage; a first FET comprising a first drain, a first gate, and a first source, wherein the first gate is connected to the output of the op amp; a resistor comprising a first terminal and a second terminal, wherein the first terminal is electrical connected to the first source and to the second input of the op amp, and wherein the second terminal is electrically connected to a ground; a voltage supply; a second FET comprising a second drain, a second gate, and a second source, wherein the second drain is electrically connected to a voltage source; a third FET comprising a third drain, a third gate, and a third source, wherein the third drain is electrically connected to the voltage source, and wherein the third gate is electrically connected to the second gate, the second source, and to the first drain; an output terminal of the current generator circuit, wherein the output terminal of the current generator circuit is electrically connected to the third source; and an IDAC electrically connected to the output terminal of the current generator circuitry.

[0006] In some embodiments, the IDAC is configured to generate an IDAC current based at least on a reference current generated by the current generator circuitry.

[0007] In some embodiments, the current generator circuitry is configured to generate the reference current based on the resistor, and wherein a change in the resistance of the resistor based on a change in temperature is associated with a change in the reference current.

[0008] In some embodiments, the IDAC is further configured to generate the IDAC current based on a selection signal comprising a plurality of bits.

[0009] In some embodiments, the plurality of bits comprises 8 bits, and wherein the 8 bits specify to the IDAC a number of current increments to implement in the IDAC current.

[0010] In some embodiments, the apparatus further comprises PWM circuitry configured to generate a driver signal based at least on the IDAC current.

[0011] In some embodiments, the apparatus further comprises a SPAD array configured to be driven based at least on the driver signal.

[0012] In some embodiments, the second FET and the third FET are PMOSFETs.

[0013] In some embodiments, the first FET is an NMOSFET.

[0014] In some embodiments, the apparatus is one of a mobile phone, laptop, or augmented reality device.

[0015] In accordance with some embodiments of the present disclosure, an example method is provided. The method may comprise: generating a first reference current with a current generator circuitry, wherein the current generator circuitry comprises: an op amp comprising a first input, a second input, and an output, wherein the first input is electrically connected to a reference voltage; a first FET comprising a first drain, a first gate, and a first source, wherein the first gate is connected to the output of the op amp; a resistor comprising a first terminal and a second terminal,

wherein the first terminal is electrical connected to the first source and to the second input of the op amp, and wherein the second terminal is electrically connected to a ground; a voltage supply; a second FET comprising a second drain, a second gate, and a second source, wherein the second drain is electrically connected to a voltage source; a third FET comprising a third drain, a third gate, and a third source, wherein the third drain is electrically connected to the voltage source, and wherein the third gate is electrically connected to the second gate, the second source, and to the first drain; an output terminal of the current generator circuit, wherein the output terminal of the current generator circuit is electrically connected to the third source; generating a first IDAC current with an IDAC at a first time based on the first reference current, wherein the IDAC is electrically connected to the output terminal of the current generator circuit; generating an adjusted reference current with the current generator circuitry at a second time based on a change in temperature; generating an adjusted IDAC current with the IDAC at a second time based on the adjusted reference current.

[0016] In some embodiments, generating the first reference current is based on the resistor, and wherein generating the adjusted reference current is based on a change in temperature associated with a change in the reference current.

[0017] In some embodiments, generating the first IDAC current is based on a selection signal comprising a plurality of bits, and wherein generating the adjusted IDAC current is based on the selection signal.

[0018] In some embodiments, the plurality of bits comprises 8 bits, and wherein the 8 bits specify to the IDAC a plurality of current increments to implement in the IDAC current.

[0019] In some embodiments, each voltage increment of the plurality of increments is associated with a linear current change.

[0020] In some embodiments, the method further comprises generating a driver signal with a PWM circuitry based at least on the IDAC current.

[0021] In some embodiments, the method further comprises driving a SPAD array with the driver signal.

[0022] In some embodiments, wherein the second FET and the third FET are PMOSFETs.

[0023] In some embodiments, the first FET is an NMOSFET.

[0024] In some embodiments, the current generator circuitry and the IDAC are in one of a mobile phone, laptop, or augmented reality device.

[0025] The above summary is provided merely for purposes of summarizing some example embodiments to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be appreciated that the above-described embodiments are merely examples and should not be construed to narrow the scope or spirit of the disclosure in any way. It will also be appreciated that the scope of the disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described below.

Description

BRIEF SUMMARY OF THE DRAWINGS

[0026] Having thus described certain example embodiments of the present disclosure in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0027] FIG. 1 illustrates an example circuit diagram associated with IDAC circuitry in accordance with one or more embodiments of the present disclosure;

[0028] FIG. 2 illustrates an example circuit diagram associated with driver circuitry in accordance with one or more embodiments of the present disclosure;

[0029] FIG. 3 illustrates an example circuit diagram associated with error circuitry in accordance

with one or more embodiments of the present disclosure;

[0030] FIG. 4 illustrates an example circuit diagram associated with driver output circuitry in accordance with one or more embodiments of the present disclosure;

[0031] FIG. 5 illustrates an example diagram associated with operations for operating an exemplary system and/or apparatus in accordance with one or more embodiments of the present disclosure; and

[0032] FIG. 6 illustrates an example block diagram of a device in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0033] Some embodiments of the present disclosure will now be described more fully herein with reference to the accompanying drawings, in which some, but not all, embodiments of the disclosure are shown. Indeed, various embodiments of the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout.

[0034] As used herein, the term “comprising” means including but not limited to and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of.

[0035] The phrases “in various embodiments,” “in one embodiment,” “according to one embodiment,” “in some embodiments,” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0036] The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0037] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments or it may be excluded.

[0038] The use of the term “circuitry” as used herein with respect to components of a system or an apparatus should be understood to include particular hardware configured to perform the functions associated with the particular circuitry as described herein. The term “circuitry” should be understood broadly to include hardware and, in some embodiments, software for configuring the hardware. For example, in some embodiments, “circuitry” may include processing circuitry, communications circuitry, input/output circuitry, and the like. In some embodiments, other elements may provide or supplement the functionality of particular circuitry.

Overview

[0039] Various embodiments of the present disclosure are directed to improved current generators for IDAC corner independence. For example, in various embodiments the current generator makes a reference current temperature and corner dependent so that an IDAC generates a voltage independent of corners and temperature.

[0040] A current digital-to-analog converter (IDAC) may generate a current output and/or voltage output that is based on at least an input reference signal (e.g., reference current or reference voltage) and a select signal. A reference current may also be described as a bias current. The select signal may be a plurality of bits, which may be referred to as a codeword or word. In various embodiments, the select signal may be 8 bits that are provided to the IDAC. In various embodiments the select signal may be fewer than 8 bits or greater than 8 bits. The IDAC may

generate an output that changes in voltage and/or current based on the select signal. For example, if an IDAC's voltage output range is from 16 to 25V, an 8 bit select signal may be used to determine how the output voltage is incremented by stepping up the voltage by a plurality of steps. A step may shift the output voltage from a low voltage value (e.g., 16V) to a high voltage value (e.g., 26V) at regular step intervals (e.g., 50 mV). Similarly, a current output may be incremented based on a select signal. Of the bits of the select signal, a least significant bit (LSB) determines the last step to be implemented by the IDAC and, thus, the output voltage and/or current. However, the IDAC out may also move due to corner dependencies of process and/or temperature. Changes in process and/or temperature result a variation of the IDAC output voltage. For example, a temperature variation may have a voltage variation of 1.2 to 2 mV in the output of the IDAC per degree C. variation. Embodiments of the present disclosure compensate for such variations in voltage. Additionally or alternatively, process corner variations include manufacturing corners to be compensated for. As described herein, the present disclosure compensates for process corners in conjunction with temperature variations. Such compensations provide, among other things, improvements by omitting the requirement of performing one or more trim processes (e.g., additional bias operations) to set the voltage increments per IDAC code to compensate for manufacturing corners. Thus the circuitry of the present disclosure may allow for and control the feedback of an application in a linear manner.

[0041] An IDAC is one component of an apparatus and/or system. The current generator circuitry and the IDAC circuitry may be referred to as a boost controller circuit or a part of a boost controller circuit. In various embodiments of the present disclosure, a high voltage value of an IDAC may be determined as a product of a reference current to the IDAC and a resistance value of a current generator circuitry. In various embodiments, the IDAC output of an IDAC current may be provided to a current divider to generate a specific voltage for a divided voltage signal. The divided voltage signal may then be input into additional circuitry depending on the application.

[0042] For example, embodiments of the present disclosure include the divided voltage signal input to additional circuitry that includes PWM circuitry and a driver to generate an output signal that may be used by or used to drive a single-photon avalanche diode (SPAD) array. Various embodiments of such circuitry are part of laptop, mobile phones, mobile devices configured for, among other things, augmented reality, and the like that may use SPAD arrays in time-of-flight (ToF) sensors to generate a light signal that is received to determine a distance by the ToF sensor. Alternatively or additionally, the SPAD arrays may be use for LIDAR functionality. These sensors benefit from the voltage generator circuitry output compensating for temperature and/or process that allows for linear outputs to the IDAC circuitry and subsequent circuitry to improve reliability and efficiency in operations. For example, the breakdown voltage of a SPAD varies with temperature. Temperature changes may be monitored with a temperature sensor. Various embodiments may also compensate the output of the IDAC circuitry through control of the voltage divider circuitry based on a temperature sensor signal turning on and/or off switches to reconfigure the resistance(s) of the voltage divider circuitry, which allows for dynamic compensation of the boost controller circuit.

[0043] Various embodiments herein may include a boost controller circuit with a voltage output that ranges between a high voltage (e.g., 25V) and a low voltage (e.g., 0V or 15V). The high voltage may be set with resistor circuitry described herein. The resistor circuitry may allow for tuning to provide more or less current. With the embodiments of the IDAC described herein, this tuning may compensate for temperature and/or process variations.

Exemplary Systems, Apparatuses, and Methods

[0044] Embodiments of the present disclosure herein include systems, apparatuses, and methods for current generator for an IDAC described herein may be implemented in various embodiments.

[0045] FIG. 1 illustrates an example circuit diagram associated with IDAC circuitry in accordance with one or more embodiments of the present disclosure. FIG. 1 illustrates an IDAC circuitry 100,

which may include a current generator circuitry **110**. While current generator circuitry **110** is illustrated in IDAC circuitry **100**, the current generator circuitry **110** may be located external to the IDAC circuitry **100**.

[0046] The IDAC circuitry **100** may receive a reference voltage at a first IDAC circuitry input terminal **102**, receive a selection signal at the second IDAC circuitry input terminal **104**, and may output an IDAC current **170** at IDAC circuitry output terminal **106**.

[0047] The current generator circuitry **110** may generate a reference current **114** based on the reference voltage, which may provide to the current generator circuitry via the second IDAC circuitry input terminal **104**. In various embodiments, the reference current **114** may be provided to a current generator circuitry output terminal **112**. The IDAC **120** may be electrically connected to the current generator circuitry output terminal **112** to receive the reference current **114** as an input to the IDAC **120**.

[0048] The IDAC **120** may generate the IDAC current **170** based at least on the reference current **114** and the selection signal. The selection signal may be comprised of a plurality of bits. In various embodiments, an 8-bit selection signal may be used. The 8 bits of the selection signal may specify how many of a plurality of increments or steps to increment a current and/or a voltage by in the IDAC **120**. Thus the IDAC **120** may generate the IDAC current **170**. In various embodiments, a change in increment may increase the current and/or voltage linearly. Alternatively or additionally, a change in increment may increase the current and/or voltage non-linearly.

[0049] The current generator circuitry **110** may include a plurality of electrical components. For example, the current generator circuitry **110** may include one or more electrical connections to a voltage rail **120**, which may also be referred to as a voltage source. The voltage rail **120** is electrically connected to a first FET **152** and a second FET **154**.

[0050] The first FET **152** may be a PMOSFET with a drain, gate, and source. The first FET **152** may be electrically connected to the voltage rail **120** at the drain of the first FET **152**.

[0051] The second FET **154** may be a PMOSFET with a drain, gate, and source. The second FET **154** may be electrically connected to the voltage rail **120** at the drain of the second FET **154**. The source of the second FET **154** may be electrically connected to the current generator circuitry output terminal **112**, which may be where the reference current **114** is output.

[0052] The gate of the first FET **152** and the gate of the second FET **154** may be electrically connected to each other, to the source of the first FET **152**, and to a drain of a third FET **156**.

[0053] The third FET **156** may be an NMOSFET with a drain, gate, and source. The gate of the third FET **156** may be electrically connected to the output of an op amp **140**. A first input (e.g., + terminal) of the op amp **140** may be electrically connected to the first IDAC circuitry input terminal **102** to receive a voltage. The source of the third FET **156** may be electrically connected to a second input (e.g., - terminal) of the op amp **140** as well as a first terminal of a resistor **130**. The resistor **130**, at a second terminal, may be electrically connected to a ground **132**. It will be appreciated that while a first FET **152**, a second FET **154**, and a third FET **156** are illustrated in FIG. **1**, various embodiments may include more than 3 FETs. For example, various embodiments may include additional FETs such that the current mirrors of the first FET **152** and the second FET **154** are cascaded with the additional FETs.

[0054] In operation, the resistor **130** varies with temperature and/or process. The variation in the resistor **130** causes a variation in the reference current **114**. The variation in resistor **130** causes a change in the current into resistor **130**. The voltage at the first terminal of resistor **130** remains constant. The op amp **140** and the third FET **156** cause the voltage across resistor **130** to the voltage on first IDAC circuitry input terminal **102**. The current varies inversely with the resistance of resistor **130**. This results in a change with the variations to the current to the first FET **152** and second FET **154** from the voltage rail **120**. This then changes reference current **114** to the IDAC **120**, which will change the IDAC current **170**.

[0055] The variation in the reference current **114** provides a variation or modulation in an input to

the IDAC **120**. This variation along with the processing of the LSB of the selection signal generates an IDAC current **170**. A variation in the resistance of resistor **130**, such as based on variations in temperature and/or process, results in a variation of the IDAC current **170**.

[0056] For example, a variation of temperature and/or process may cause a variation in the reference current **114** that compensates for a plus or minus variation $80\ \mu\text{A}$ in the reference current **114**. As voltage is related to current and the resistance, and as the voltage at the first terminal of resistor **130** is caused to be the voltage at the first IDAC circuitry input terminal **102**, the variation in the resistance varies the IDAC current **170** from the IDAC **120**. In various embodiments, when. In various embodiments, the voltage at the first IDAC circuitry input terminal **102** may be referred to as a bandgap voltage. When the IDAC current **170** is provided to a matching resistor, the voltage will not change with corners.

[0057] FIG. **2** illustrates an example circuit diagram associated with driver circuitry in accordance with one or more embodiments of the present disclosure. The driver circuitry **210** may include, among other things, the IDAC circuitry **100**. The IDAC current **170** of the IDAC circuitry **100**, which is varied or compensated with temperature and/or process variations, may be used by the driver circuitry **210** to generate a driver signal **290** that may be provided to a driver circuitry output terminal **218**. A driver output circuitry **220** may be electrically connected to the driver circuitry output terminal **218** and may control the application of the driver signal **290** to, for example, a SPAD array (not illustrated) that may be electrically connected to an output terminal **222**. Thus the driver signal **290** may be used to drive the SPAD array, such as in ToF applications, augmented reality applications, and the like. The output terminal **222** may also be electrically connected to a driver circuitry input terminal **212** to provide a feedback signal **224**. The feedback signal **224** may be a voltage signal that may be used by the driver circuitry **210** in generating the driver signal **290**.

[0058] While the driver output circuitry **220** is illustrated separate from the driver circuitry **210**, it will be appreciated the driver output circuitry **220** may be located in or adjacent to the driver circuitry **210**.

[0059] The driver circuitry **210** may include, among other things, resistor circuitry **230**, error circuitry **250**, and PWM circuitry **260**. The driver circuitry **210** may include a plurality of input terminals (e.g., **212**, **214**) and one or more output terminal (e.g., **218**).

[0060] The resistor circuitry **230** may include a plurality of resistors, including a first or top resistor **232**, a second or bottom resistor **234**, and a ground **240**. It will be appreciated that each of the first resistor **232** and also the second resistor **234** may be comprised of, among other things, two or more resistors in a plurality of configurations.

[0061] The first resistor **232** may include a first terminal and a second terminal. The second resistor **234** may also include a first terminal and a second terminal.

[0062] The first terminal of the first resistor **232** may be electrically connected to the driver circuitry input terminal **212** to receive the feedback signal **224**. The second terminal of the first resistor **232** may be electrically connected to the first terminal of the second resistor **234**, to the IDAC circuitry output terminal **106** to receive the IDAC current **170**, to the resistor circuitry output terminal **236** to provide the divided voltage signal **270**. The second terminal of the second resistor **234** may be electrically connected to a ground **240**.

[0063] The divided voltage signal **270** may be determined based on the feedback signal **224**, the IDAC current **170**, and the ratio of resistances of the first resistor **232** and the second resistor **234** of the resistance circuitry **230**. The IDAC current **170** from the IDAC **120** will bias the second resistor **234**. With the feedback loop into the op amp **140**, the IDAC current **170** will pull the input nodes of the op amp **140** equal to a reference voltage (e.g., $0.9\ \text{V}$). This causes the temperature and/or process variations in the voltage divider circuitry **230** to be cancelled by the resistance ratio.

[0064] The resistance of the first resistor **232** and of the second resistor **234** may be chosen to be multiples of resistor **130**. For example, the multiple **M1** may be a multiple of the resistance of resistor **130** that is the resistance of the first resistor **232**. The multiple **M2** may be a multiple of the

resistance of resistor **130** that is the resistance of the second resistor **234**. As such, the following equations provide for values of certain voltages.

$$VFB = VREF_ERR + R1 * (I2 - IDAC)$$

$VFB = VREF_ERR * (1 + R1/R2) - N * R1/R * VREF_IDAC$ [0065] Setting $VREF_ERR$ equal to $VREF_IDAC$ may allow for both to be written as $VREF$, which allows for VFB to be:

$VFB = VREF * (1 + R1/R2 - N * R1/R)$ [0066] VFB =feedback voltage **224** (which is also the output voltage at terminal **222**) [0067] $VREF_ERR$ =a first reference voltage, such as input at driver circuitry input terminal **214** [0068] $VREF_IDAC$ =a second reference voltage, such as input at IDAC circuitry input terminal **102** [0069] $IDAC$ =reference current **170** [0070] $I2$ =current through resistor **R2** [0071] R =resistance of resistor **130** [0072] $R1 = M1 * R$ =resistance of resistor **232** [0073] $R2 = M2 * R$ =resistance of resistor **234** [0074] N =IDAC settings from selection signal [0075] For example, a first configuration of the resistor circuitry **230** may include a voltage ladder with a plurality of resistors and selection switches or taps. The selection switches may be used to open or close electrically connections to the plurality of resistors to vary the voltage and/or current that may be provided at the resistor circuitry output terminal **236** for the divided voltage signal **270**. In various embodiments the selection switches and/or taps may be open or closed to select a voltage based on a control signal received, which may be based on a temperature measurement from one or more temperature sensors. In such embodiments the voltage may be varied according to, among other things, temperature measurements.

[0076] The divided voltage signal **270** may be provided to the error circuitry **250** to generate an error voltage **280**. The error circuitry **250** is described further herein.

[0077] The error voltage **280** is provided to the PWM circuitry **260**. The PWM circuitry **260** may include a PWM modulator, a pulse, and one or more drivers. The PWM modulator may use a clock signal and the error voltage **280** to generate a PWM signal **290**. The PWM signal may be a shaped signal. The PWM signal **290** may be provided to a pulse shaper to shape one or more pulses to provide to one or more drivers. The one

[0078] FIG. **3** illustrates an example circuit diagram associated with error circuitry in accordance with one or more embodiments of the present disclosure. The error circuitry **250** may include an op amp **310**, a plurality of resistors (e.g., **312**, **314**, and **316**, and plurality of resistors (e.g., **322**, **324**, and **326**). The error circuitry **250** may include a first input terminal **252**, a second input terminal **254**, and an output terminal **256**. The first input terminal **252** may be electrically connected to other circuitry (e.g., driver circuitry input terminal **214**) to provide a reference voltage to the op amp **310**. The second input terminal **254** may be electrically connected to resistor circuitry output terminal **236** to provide the divided voltage signal **270** the plurality of resistors, plurality of capacitors, and then the op amp **310**. The output terminal **256** may provide the error voltage **280** to additional circuitry (e.g., PWM circuitry **260**). It will be appreciated that the error circuitry may include more or less electrical components than illustrated in FIG. **3**.

[0079] FIG. **4** illustrates an example circuit diagram associated with driver output circuitry in accordance with one or more embodiments of the present disclosure. The driver output circuitry **220** may include a voltage rail **410**, which may be a voltage source. It may also include a FET **430**, a diode **440**, a capacitor **450**, and a plurality of grounds **460A**, **460B**. The plurality of grounds may also be a single ground. The driver output circuitry **220** may include an input terminal **218** to receive a driver signal **290** and an output terminal **222**. The feedback signal **224** may be electrically connected to the output terminal **222**. In various embodiments, the output terminal **222** may be electrically connected to one or more additional circuitries, such as to a SPAD array. The driver output circuitry **220** operate as a gate to control the transmission of the driver signal **290** from the input terminal **218** to the output terminal **222**.

[0080] FIG. 5 illustrates an example diagram associated with operations for operating an exemplary system and/or apparatus in accordance with one or more embodiments of the present disclosure.

[0081] At operation **502**, providing current generator circuitry, such as described herein.

Additionally, the current generator circuitry **110** may be provided with additional circuitry described herein.

[0082] At operation **504**, generating reference current with current generator circuitry. The current generator circuitry **110** may be used to generate a reference current **114**.

[0083] At operation **506**, generating IDAC current based at least on the reference current. The IDAC **120** may be used to generate an IDAC current **170** based at least on the reference current **114**. For example, the IDAC **120** may also use a selection signal to generate the IDAC current **170**.

[0084] At operation **508**, generating adjusted reference current based at least on a temperature change. A variation in temperature from time **1** to time **2** may lead to a temperature change. The resistor **130** of the current generator circuitry **110** may vary with the change in temperature. This may, as described herein, generate a reference current **114** that varies from the reference current **114** generated at time **1**. The reference current **114** at time **2** that includes the variation may be referred to as an adjusted reference current.

[0085] At operation **510**, generate adjusted IDAC current based at least on adjusted reference current. With the adjusted reference current, the IDAC **120** may generate an adjusted IDAC current at time **2**.

[0086] At operation **512**, generate an adjusted driver signal. Various embodiments may include PWM circuitry **260** that may be used to generate a driver signal **290**. The variation of the IDAC current **170** may be associated with a variation in the driver signal **290**. At a first time the PWM circuitry **260** may generate a first driver signal **290** and at time **2**, after a variation in temperature, the PWM circuitry **260** may generate an adjusted driver signal **290** that varies based at least on the adjusted reference current generated by the current generator circuitry **110**.

[0087] At operation **514**, biasing SPAD array with adjusted driver signal. In various embodiments the driver signal **290** may be provided to a SPAD array. For example, the driver signal **290** may be transmitted through driver output circuitry **220** and a SPAD array may be electrically connected to output terminal **222**. The driver signal **290** may be adjusted based on the variation in temperature and then the adjusted driver signal **290** may be provided to the SPAD array for use in application, including but not limited to applications described herein.

[0088] FIG. 6 illustrates an example block diagram of a device in accordance with one or more embodiments of the present disclosure. Exemplary embodiments of the device **600** may include, for example, mobile phone, laptop, augmented reality device, etc. The device **600** illustrated includes a processor **602**, memory **604**, communications circuitry **606**, and input/output circuitry **608**, and sensor circuitry **610**, which may all be connected via a bus **612**.

[0089] The processor **602**, although illustrated as a single block, may be comprised of a plurality of components and/or processor circuitry. The processor **602** may be implemented as, for example, various components comprising one or a plurality of microprocessors with accompanying digital signal processors; one or a plurality of processors without accompanying digital signal processors; one or a plurality of coprocessors; one or a plurality of multi-core processors; processing circuits; and various other processing elements. The processor may include integrated circuits, such as ASICs, FPGAs, systems-on-a-chip (SoC), or combinations thereof. In various embodiments, the processor **602** may be configured to execute applications, instructions, and/or programs stored in the processor **602**, memory **604**, or otherwise accessible to the processor **602**. When executed by the processor **602**, these applications, instructions, and/or programs may enable the execution of one or a plurality of the operations and/or functions described herein. Regardless of whether it is configured by hardware, firmware/software methods, or a combination thereof, the processor **602** may comprise entities capable of executing operations and/or functions according to the embodiments of the present disclosure when correspondingly configured.

[0090] The memory **604** may comprise, for example, a volatile memory, a non-volatile memory, or a certain combination thereof. Although illustrated as a single block, the memory **604** may comprise a plurality of memory components. In various embodiments, the memory **604** may comprise, for example, a random access memory, a cache memory, a flash memory, a hard disk, a circuit configured to store information, or a combination thereof. The memory **604** may be configured to write or store data, information, application programs, instructions, etc. so that the processor **602** may execute various operations and/or functions according to the embodiments of the present disclosure. For example, in at least some embodiments, a memory **604** may be configured to buffer or cache data for processing by the processor **602**. Additionally or alternatively, in at least some embodiments, the memory **604** may be configured to store program instructions for execution by the processor **602**. The memory **604** may store information in the form of static and/or dynamic information. When the operations and/or functions are executed, the stored information may be stored and/or used by the processor **602**.

[0091] The communication circuitry **606** may be implemented as a circuit, hardware, computer program product, or a combination thereof, which is configured to receive and/or transmit data from/to another component or apparatus. The computer program product may comprise computer-readable program instructions stored on a computer-readable medium (e.g., memory **604**) and executed by a processor **602**. In various embodiments, the communication circuitry **606** (as with other components discussed herein) may be at least partially implemented as part of the processor **602** or otherwise controlled by the processor **602**. The communication circuitry **606** may communicate with the processor **602**, for example, through a bus **612**. Such a bus **612** may connect to the processor **602**, and it may also connect to one or more other components of the processor **602**. The communication circuitry **606** may be comprised of, for example, transmitters, receivers, transceivers, network interface cards and/or supporting hardware and/or firmware/software, and may be used for establishing communication with another component(s), apparatus(es), and/or system(s). The communication circuitry **606** may be configured to receive and/or transmit data that may be stored by, for example, the memory **604** by using one or more protocols that can be used for communication between components, apparatuses, and/or systems.

[0092] In various embodiments, the communication circuitry **606** may convert, transform, and/or package data into data packets and/or data objects to be transmitted and/or convert, transform, and/or unpack data received, such as from a first protocol to a second protocol, from a first data type to a second data type, from an analog signal to a digital signal, from a digital signal to an analog signal, or the like. The communication circuitry **606** may additionally, or alternatively, communicate with the processor **602**, the memory **604**, the input/output circuitry **608**, and/or the sensor circuitry **610**, such as through a bus **612**.

[0093] The input/output circuitry **608** may communicate with the processor **602** to receive instructions input by an operator and/or to provide audible, visual, mechanical, or other outputs to an operator. The input/output circuitry **608** may comprise supporting devices, such as a keyboard, a mouse, a user interface, a display, a touch screen display, lights (e.g., warning lights), indicators, speakers, and/or other input/output mechanisms. The input/output circuitry **608** may comprise one or more interfaces to which supporting devices may be connected. In various embodiments, aspects of the input/output circuitry **608** may be implemented on a device used by the operator to communicate with the processor **602**. The input/output circuitry **608** may communicate with the memory **604**, the communication circuitry **606**, the sensor circuitry **610**, and/or any other component, for example, through a bus **612**.

[0094] The sensor circuitry **610** may include the driver circuitry **210** and driver output circuitry **220** as well as sensors (e.g., a SPAD array, etc.). The sensor circuitry **610** may be implemented as circuitry described herein as well as additional hardware, computer program products, or a combination thereof, which is configured to perform one or more operations and/or function, such as those described herein.

[0095] It should be readily appreciated that the embodiments of the systems, apparatuses, and methods and related systems and apparatuses described herein may be configured in various additional and alternative manners in addition to those expressly described herein. For example an operation may include additional operations and/or omit one or more operations described herein.

CONCLUSION

[0096] Operations and/or functions of the present disclosure have been described herein, such as in flowcharts. As will be appreciated, computer program instructions may be loaded onto a computer or other programmable apparatus (e.g., hardware) to produce a machine, such that the resulting computer or other programmable apparatus implements the operations and/or functions described in the flowchart blocks herein. These computer program instructions may also be stored in a computer-readable memory that may direct a computer, processor, or other programmable apparatus to operate and/or function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture, the execution of which implements the operations and/or functions described in the flowchart blocks. The computer program instructions may also be loaded onto a computer, processor, or other programmable apparatus to cause a series of operations to be performed on the computer, processor, or other programmable apparatus to produce a computer-implemented process such that the instructions executed on the computer, processor, or other programmable apparatus provide operations for implementing the functions and/or operations specified in the flowchart blocks. The flowchart blocks support combinations of means for performing the specified operations and/or functions and combinations of operations and/or functions for performing the specified operations and/or functions. It will be understood that one or more blocks of the flowcharts, and combinations of blocks in the flowcharts, can be implemented by special purpose hardware-based computer systems which perform the specified operations and/or functions, or combinations of special purpose hardware with computer instructions.

[0097] While this specification contains many specific embodiments and implementation details, these should not be construed as limitations on the scope of any disclosures or of what may be claimed, but rather as descriptions of features specific to particular embodiments of particular disclosures. Certain features that are described herein in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0098] While operations and/or functions are illustrated in the drawings in a particular order, this should not be understood as requiring that such operations and/or functions be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, operations and/or functions in alternative ordering may be advantageous. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results. Thus, while particular embodiments of the subject matter have been described, other embodiments are within the scope of the following claims.

[0099] While this detailed description has set forth some embodiments of the present invention, the appended claims cover other embodiments of the present invention which differ from the described embodiments according to various modifications and improvements.

[0100] Within the appended claims, unless the specific term “means for” or “step for” is used within a given claim, it is not intended that the claim be interpreted under 35 U.S.C. § 112, paragraph 6.

Claims

1. An apparatus comprising: a current generator circuitry comprising: an op amp comprising a first input, a second input, and an output, wherein the first input is electrically connected to a reference voltage; a first FET comprising a first drain, a first gate, and a first source, wherein the first gate is connected to the output of the op amp; a resistor comprising a first terminal and a second terminal, wherein the first terminal is electrical connected to the first source and to the second input of the op amp, and wherein the second terminal is electrically connected to a ground; a voltage supply; a second FET comprising a second drain, a second gate, and a second source, wherein the second drain is electrically connected to a voltage source; a third FET comprising a third drain, a third gate, and a third source, wherein the third drain is electrically connected to the voltage source, and wherein the third gate is electrically connected to the second gate, the second source, and to the first drain; an output terminal of the current generator circuit, wherein the output terminal of the current generator circuit is electrically connected to the third source; and an IDAC electrically connected to the output terminal of the current generator circuitry.
2. The apparatus of claim 1, wherein the IDAC is configured to generate an IDAC current based at least on a reference current generated by the current generator circuitry.
3. The apparatus of claim 2, wherein the current generator circuitry is configured to generate the reference current based on the resistor, and wherein a change in the resistance of the resistor based on a change in temperature is associated with a change in the reference current.
4. The apparatus of claim 2, wherein the IDAC is further configured to generate the IDAC current based on a selection signal comprising a plurality of bits.
5. The apparatus of claim 4, wherein the plurality of bits comprise 8 bits, and wherein the 8 bits specify to the IDAC a number of current increments to implement in the IDAC current.
6. The apparatus of claim 2 further comprising PWM circuitry configured to generate a driver signal based at least on the IDAC current.
7. The apparatus of claim 6 further comprising a SPAD array configured to be driven based at least on the driver signal.
8. The apparatus of claim 1, wherein the second FET and the third FET are PMOSFETs.
9. The apparatus of claim 1, wherein the first FET is an NMOSFET.
10. The apparatus of claim 1, wherein the apparatus is one of a mobile phone, laptop, or augmented reality device.
11. A method comprising: generating a first reference current with a current generator circuitry, wherein the current generator circuitry comprises: an op amp comprising a first input, a second input, and an output, wherein the first input is electrically connected to a reference voltage; a first FET comprising a first drain, a first gate, and a first source, wherein the first gate is connected to the output of the op amp; a resistor comprising a first terminal and a second terminal, wherein the first terminal is electrical connected to the first source and to the second input of the op amp, and wherein the second terminal is electrically connected to a ground; a voltage supply; a second FET comprising a second drain, a second gate, and a second source, wherein the second drain is electrically connected to a voltage source; a third FET comprising a third drain, a third gate, and a third source, wherein the third drain is electrically connected to the voltage source, and wherein the third gate is electrically connected to the second gate, the second source, and to the first drain; an output terminal of the current generator circuit, wherein the output terminal of the current generator circuit is electrically connected to the third source; generating a first IDAC current with an IDAC at a first time based on the first reference current, wherein the IDAC is electrically connected to the output terminal of the current generator circuitry; generating an adjusted reference current with the current generator circuitry at a second time based on a change in temperature; generating an adjusted IDAC current with the IDAC at a second time based on the adjusted reference current.

- 12.** The method of claim 11, wherein generating the first reference current is based on the resistor, and wherein generating the adjusted reference current is based on a change in temperature associated with a change in the reference current.
 - 13.** The method of claim 11, wherein generating the first IDAC current is based on a selection signal comprising a plurality of bits, and wherein generating the adjusted IDAC current is based on the selection signal.
 - 14.** The method of claim 13, wherein the plurality of bits comprise 8 bits, and wherein the 8 bits specify to the IDAC a plurality of current increments to implement in the IDAC current.
 - 15.** The method of claim 14, each voltage increment of the plurality of increments is associated with a linear current change.
 - 16.** The method of claim 11 further comprising generating a driver signal with a PWM circuitry based at least on the IDAC current.
 - 17.** The method of claim 16, further comprising driving a SPAD array with the driver signal.
 - 18.** The method of claim 11, wherein the second FET and the third FET are PMOSFETs.
 - 19.** The method of claim 11, wherein the first FET is an NMOSFET.
 - 20.** The method of claim 11, wherein the current generator circuitry and the IDAC are in one of a mobile phone, laptop, or augmented reality device.
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